

A Comparative Study of Machine-Learning Classifiers for Predicting HOLC Grades

Bao Nguyen, Sherry Zhang

UNIVERSITY OF MASSACHUSETTS 1863

EMORY

Outline

01

Background

02

Methods

03

Results

04

Conclusion

05

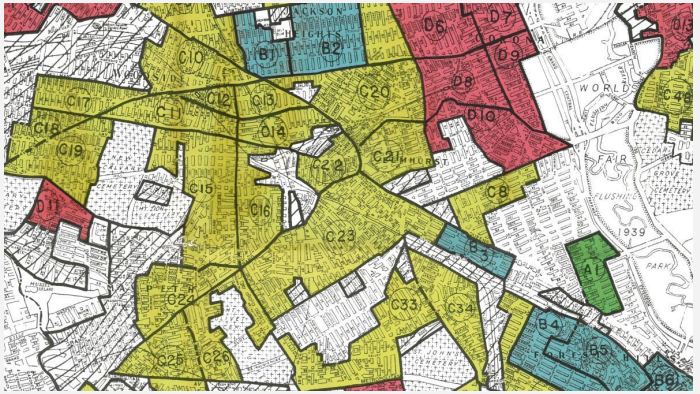
Next Steps

06

Question

Background

1. In the 1930s, the **Home Owners' Loan Corporation** (HOLC) graded neighborhoods **A–D** based on perceived credit risk
2. These grades were racially and economically biased → contributed to **redlining**
3. Decades later, HOLC grades still correlate with health disparities, pollution exposure, and wealth inequality



Research Question

How do different machine learning classifiers compare in their ability to predict HOLC grades based on socio-environmental data?

Methods

Data Cleaning

1. Raw Data: Over **400 tract-level variables** from census, pollution, SES datasets
2. Spatial Validity: Removed tracts with invalid coordinates (st_is_valid)
3. Selected variables based on pollution, demographics, COI, etc
4. Missing Values:
    a. Converted "NULL" → NA
    b. Dropped columns with **>50% missingness**
    c. Kept only **complete cases** (*n* =~6,700, subsetted into **2,047**)
5. Changing the Median Family Income to numerical
6. Variable Aggregation:
    a. Grouped 80+ related variables into 5 indices: coordinates, population, socioeconomic, pollution, Child Opportunity Index
    b. Outcome variable: **HOLC grade**
7. Deleting misleading columns: scorea, scoreb, scorec, scored, hdscore, cbsa_quartile, natl_quartile

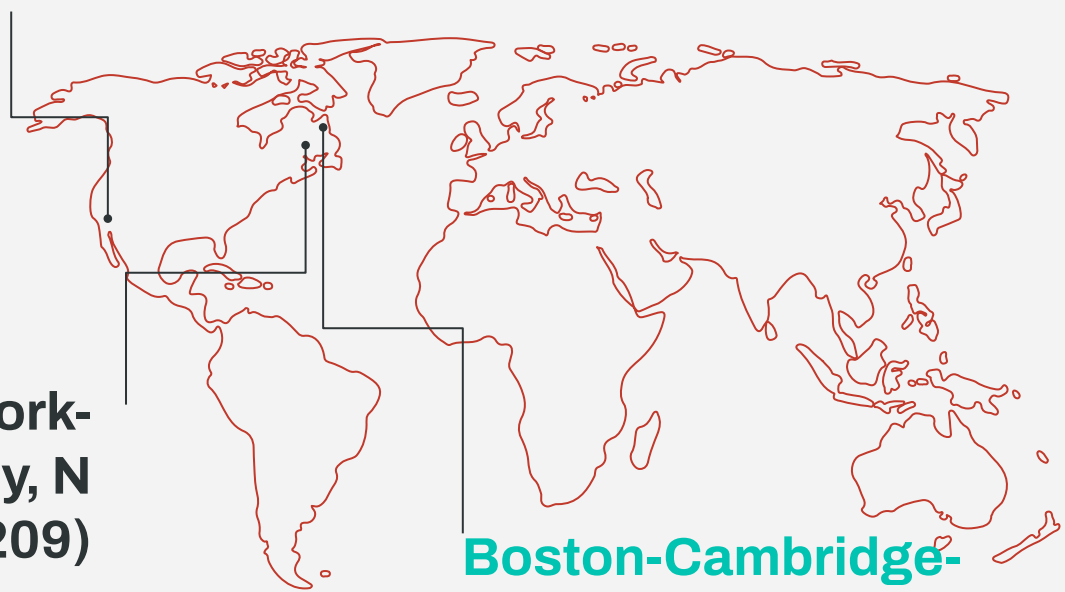
Methods

Modeling Approach

1. Data Split: training (**NY**), testing (**LA** & **Boston**)
2. Classifiers:
    a. Support Vector Machine (**SVM**), Random Forest (**RF**), k-Nearest Neighbors (**KNN**)
3. Prediction Target:
    a. HOLC class label (**A–D**) using 4 models
4. Evaluation Metrics:
    a. Accuracy
    b. F1-score (Macro)
    c. Sensitivity / Specificity
    d. Log-loss
    e. Brier score

Study Area: Train vs. Test

Los Angeles-Long Beach, CA (n=553)



New York-Newark-Jersey City, N (n=1209)

Boston-Cambridge-Newton, MA-NH (n=285)

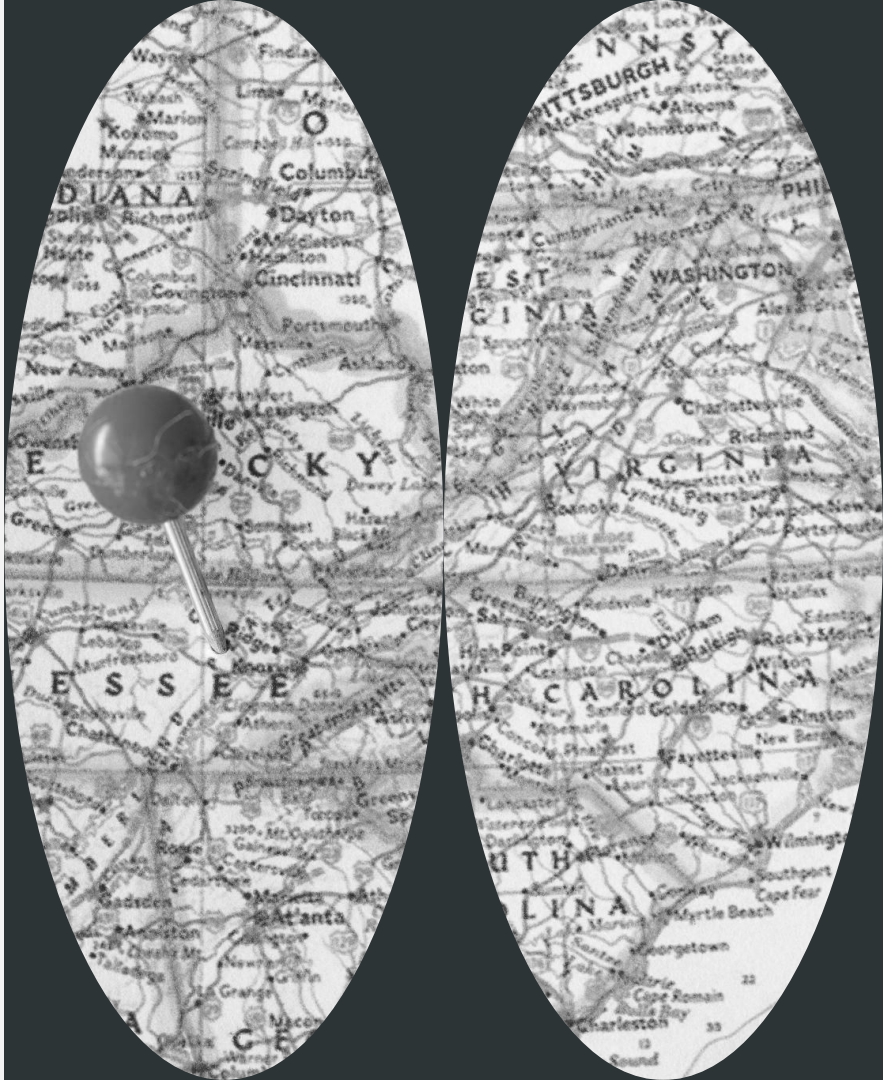
Methods

Grid Search Cross Validation Process

1. Split into k equal folds.
2. Loop over each fold i:
    a. Loop over each pair of hyperparameters from the predefined set (SVM: **cost - gamma**, KNN: **k - l**, RF: **mtry - ntree**)
        i. Use fold i as the validation set, and the remaining k–1 folds as the training set.
        ii. Train the model on the training folds.
        iii. Evaluate on the validation fold and record the metric(s).
3. Aggregate results (accuracies) across all k folds for each pair of hyperparameters for all models.
4. Pick out the pair of hyperparameter for each model that has the highest average accuracy.
5. Retrained the model based on full training data and collect assessment metrics based on testing data for Boston and Los Angeles

03

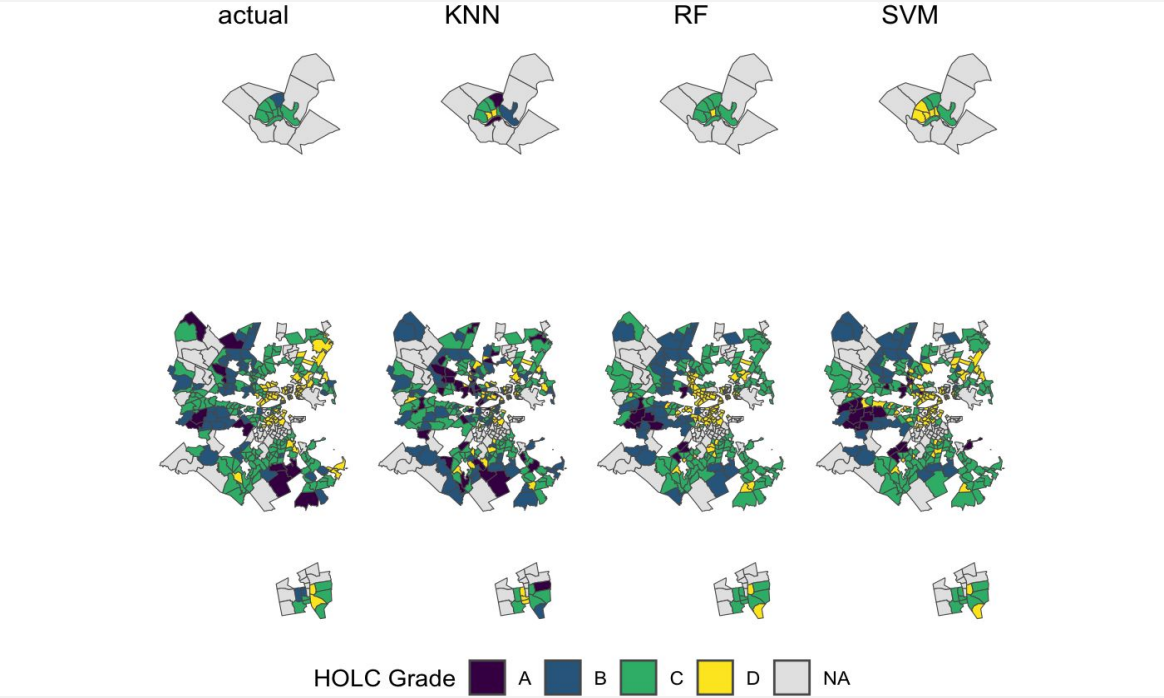
Results


Columbus
Dayton
Cincinnati
Atlanta
WASHINGTON
Charlottesville
Richmond
Roanoke
Raleigh
Wilmington
Myrtle Beach
Charleston

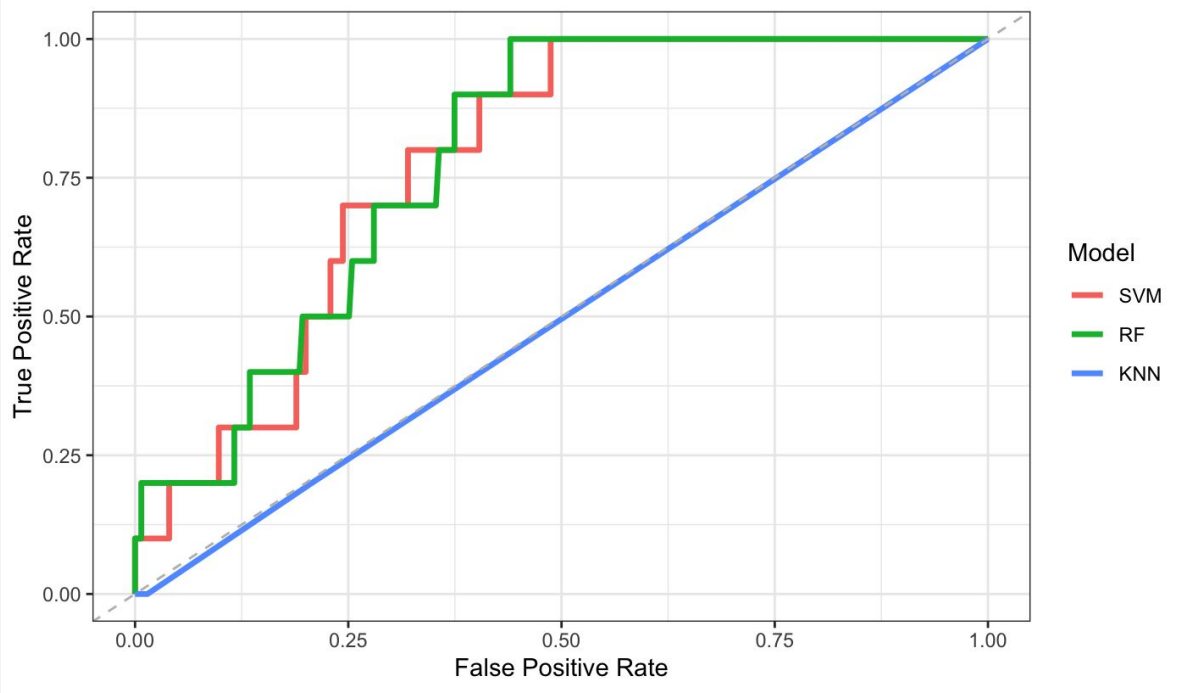
Boston: Assessment Metrics Results

Model	Accuracy	Sensitivity	Specificity	F1	Log Loss	Brier Score
SVM	0.537	0.423	0.815	0.408	2.156	0.611
RF	0.596	0.489	0.838	0.484	0.997	0.569
KNN	0.393	0.323	0.775	0.306	9.478	0.788

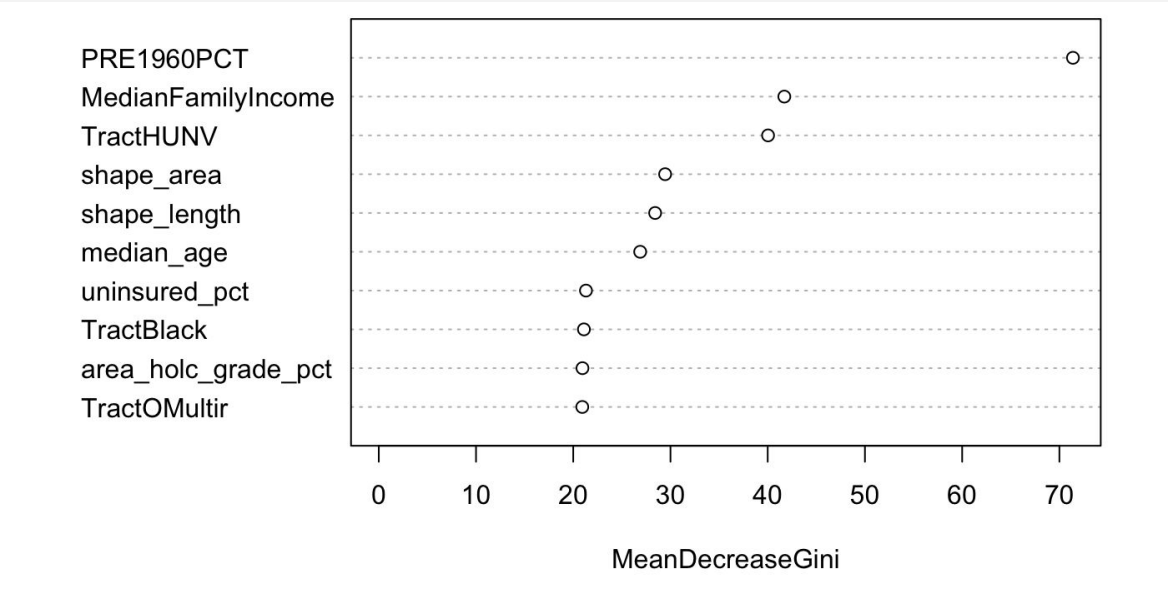
Graph Comparing Actual versus Predicted HOLC grade



ROC for Class A



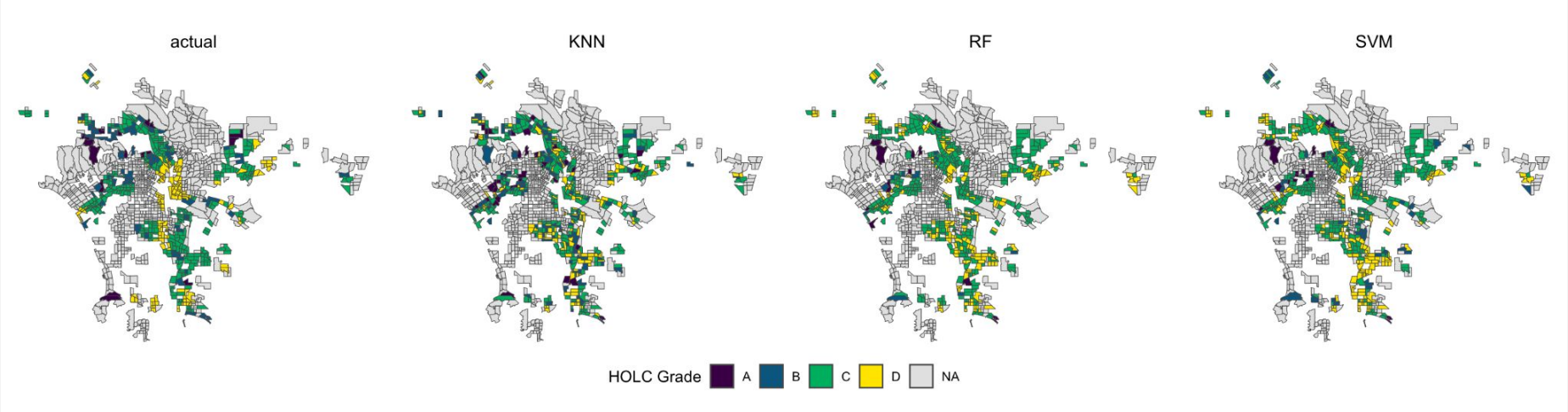
Top 10 Most Important Predictors from Random Forest



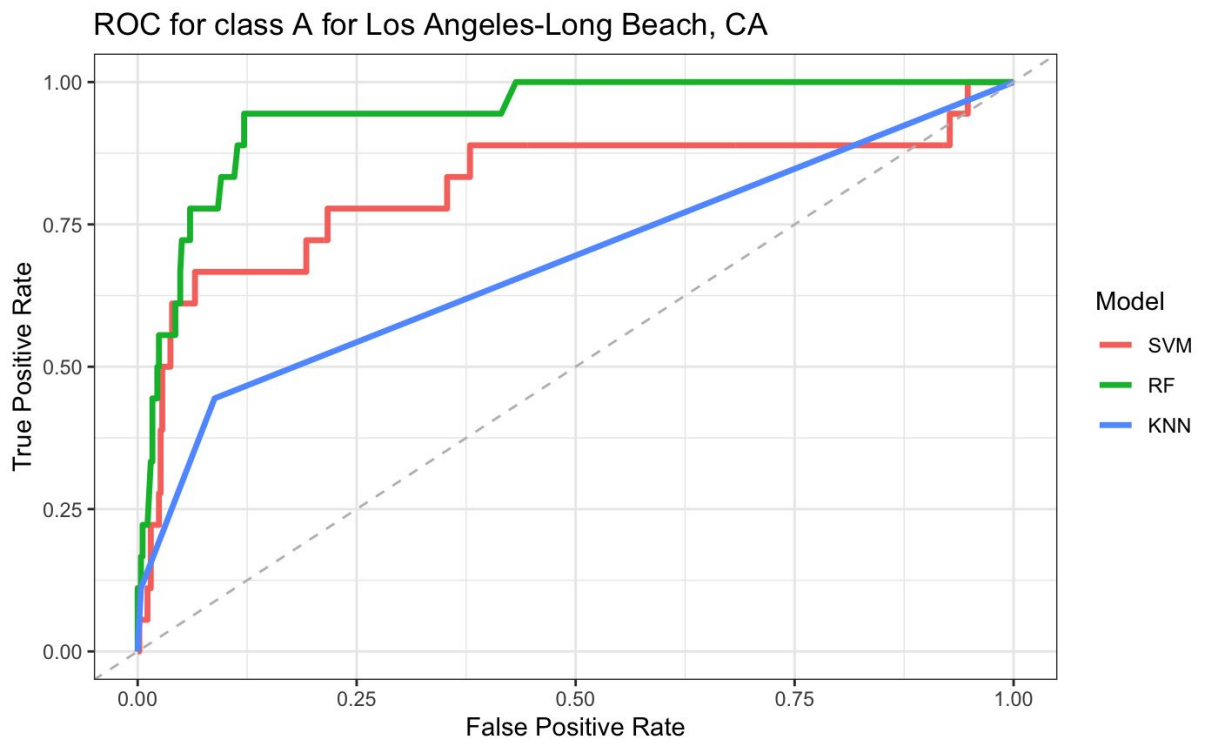
Los Angeles: Assessment Metrics Results

Model	Accuracy	Sensitivity	Specificity	F1	Log Loss	Brier Score
SVM	0.439	0.320	0.756	0.308	2.114	0.693
RF	0.447	0.327	0.746	0.316	1.01	0.609
KNN	0.389	0.324	0.757	0.299	9.461	0.798

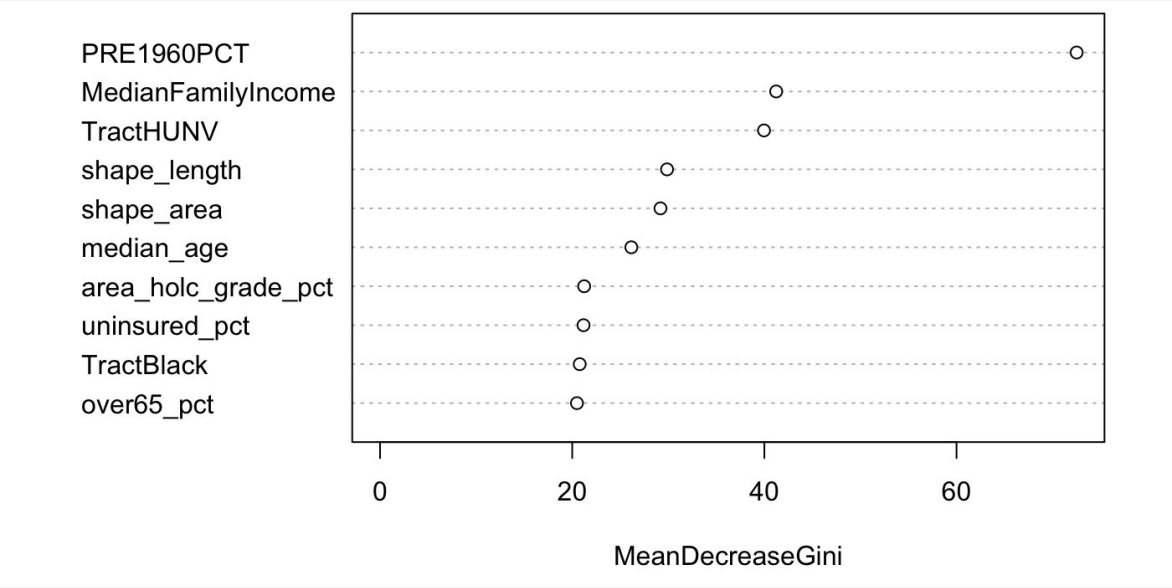
Graph Comparing Actual versus Predicted HOLC grade



ROC for Class A



Top 10 Most Important Predictors from Random Forest



Conclusion

- RF and SVM outperforms KNN on the ability to predict HOLC grade based on given dataset
- When trained on New York data, the classification models perform better on the Los Angeles test set than on the Boston test set
- Lead Paint, Median Family Income, Total count of housing units with a vehicle in tract are the top 3 most important predictor for RF

Next Steps

- Explore Advanced Models
  - Try XGBoost or deep learning to improve predictive performance and capture nonlinear patterns.
- Expand Geographic Scope
  - Apply models to other cities to test generalizability and robustness across regions.
- Add New Predictors
  - Integrate variables like school quality, healthcare access, or housing data for richer context.



Questions?

Thanks for Listening

